SINGAPORE DIMENSIONS • GRADE 8A • CHAPTER 5

Quadratic Factorisation & [POLY] Root Solver

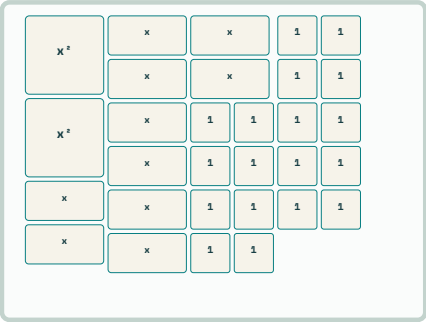
Tactile Concrete-Pictorial-Abstract (CPA) Station Guide • CCSS.MATH.HSA.REI.B.4

Student Name: _____ Date: _____ **Model: SC-32** **Check Score: ___ / 4**

STUDENT SELF-CHECK RULE (NO CHEATING):
Build with blocks and solve on paper first! Use your SC-32 calculator ONLY to verify answers on the 4-line LCD screen.

STATION LAB TUTORIAL: BENCH SCHEMATIC & CPA PROGRESSION

1. 2D Bench Blueprint



Kit: Algebra Tiles

2. Desk Setup (Concrete -> Pictorial)

Physical Bench Setup:
Arrange one x^2 tile, 5 x-tiles, and 6 unit squares into a complete rectangle: $(x+2)(x+3)$.

Sketch Model on Paper:
Draw area model rectangle labeled with dimensions $(x + 2)$ and $(x + 3)$.

3. Step-by-Step LCD Stack Trace (SC-32)

MODE: 123 Base-10 Arithmetic

1.	POLY	Step 1	Execute [POLY]
2.	1	Step 2	Execute [1]
3.	A	Step 3	Execute [A]
4.	7	Step 4	Execute [7]

EXPECTED LCD X: Roots -3, -4; Factors (x+3

1 **Factor $x^2 + 5x + 6$ via [POLY].**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-32

2 **Factor $x^2 - x - 6$ via [POLY].**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-32

3 **Difference of squares: $x^2 - 16$ via [POLY].**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-32

4 **Double root: $x^2 + 6x + 9$ via [POLY].**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-32

Station Diagnostic & Solution Key: Chapter 5

Quadratic Factorisation & [POLY] Root Solver • Manipulative Setup, Diagnostic Rubric & Verification Flow

MANIPULATIVE HARDWARE STATION INVENTORY & 3D PRINT PREPARATION

Station Kit: Algebra Studio & Factoring Stage

CAD Blueprint: algebra_tiles

Target Device: SC-32 Scientific

CAD / STL Source File:

Education/scad/algebra_factoring_parts.scad

Standard: Gridfinity 42mm Pitch / 0.6mm Inset Markings

Recommended Filaments:

Blue x^2 Flats, Green x Rods, Yellow Unit Squares (PLA)

#	Math Problem Prompt	Expected Math Answer	Self-Check Keystrokes & RPN Stack
1	**Factor $x^2 + 5x + 6$ via [POLY].** Standard: CCSS.MATH.HSA.REI.B.4	Roots -2, -3; (x+2)(x+	[POLY] [1] [A] [5] [B] [6] [C] [0] [D]
2	**Factor $x^2 - x - 6$ via [POLY].** Standard: CCSS.MATH.HSA.REI.B.4	Roots 3, -2; (x-3)(x+2	[POLY] [1] [A] [1] [+/-] [B] [6] [+/-]
3	**Difference of squares: $x^2 - 16$ via [POLY].** Standard: CCSS.MATH.HSA.REI.B.4	Roots 4, -4; (x-4)(x+4	[POLY] [1] [A] [0] [B] [1][6] [+/-] [C
4	**Double root: $x^2 + 6x + 9$ via [POLY].** Standard: CCSS.MATH.HSA.REI.B.4	Double root -3; $(x+3)^2$	[POLY] [1] [A] [6] [B] [9] [C] [0] [D]

TEACHER DIAGNOSTIC GUIDE: COMMON MISCONCEPTIONS & REMEDIATION

Key Misconception to Target During Station Observation:

Factors (x - r1)(x - r2) have opposite signs from roots r1, r2.

How the Manipulatives & Calculator Foster Self-Correction:

Students solve with pencil and physical blocks first. When they self-check on the SC-32, any discrepancy between their written answer and the calculator's 4-line LCD immediately prompts them to re-inspect their physical bench model or regrouping, turning errors into productive struggle without teacher intervention.

Station Facilitation & Pacing Tip:

Keep 3D printed trays docked on standard Gridfinity baseplates to prevent loose parts on student desks. Encourage students to articulate their CPA steps: name the block representation (Concrete), point to their drawn tape/bond model (Pictorial), and verify register values (Abstract).

CLASSROOM SCORING & STATION MASTERY RUBRIC

Exemplary (18–20 pts):	Complete, accurate manipulative sketches; exact math answers; independently verified self-check on LCD stack.
Proficient (14–17 pts):	Correct math answers; minor sketch omissions; verified on calculator with minimal teacher prompting.
Developing (10–13 pts):	Understands problem concept but made computational slips; successfully self-corrected using calculator LCD.
Intervention (<10 pts):	Relied on calculator before solving on paper, or struggled with underlying physical manipulative models.